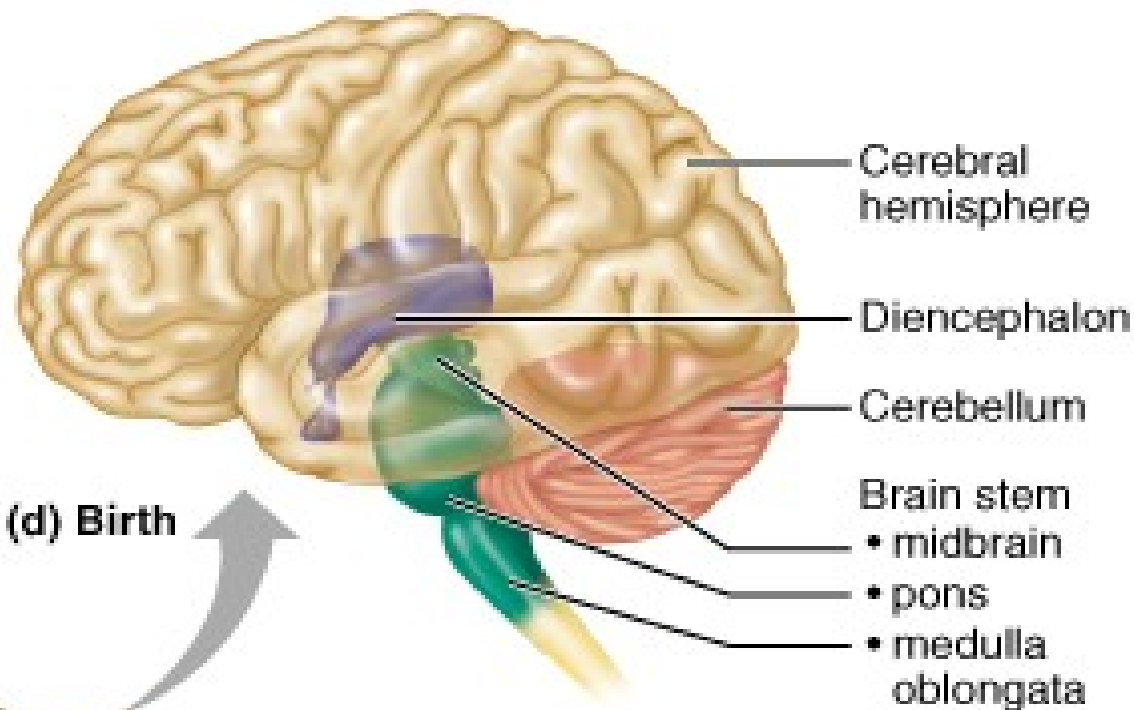


PSYCHOSIS DISORDERS



Abebaw D.

Psychotic disorders

- **Psychosis** is the term used to describe a mental state in which the individual experiences a **distortion or loss of contact with reality**, without clouding of consciousness
- A psychotic patient has a **grossly impaired sense of reality**, coupled with **emotional** and **cognitive disabilities**, which compromise **ability to function**
- Patient **talk and act in a bizarre fashion**, have **hallucination** or **strongly hold false ideas**.
- Is not aware of impairment (**lacks insight**)

Classification of Psychosis disorders

A. Schizophrenia

B. Other psychotic disorders

1. Schizophreniform
2. Schizoaffective
3. Delusional
4. Brief psychotic
5. Psychotic disorders NOS

5. Psychotic disorders NOS

- a) Post partum psychosis
- b) Psychotic Disorder Due to GMC
- c) Substance-Induced Psychotic Disorder

Schizophrenia

- **Definition:** The word **schizophrenia** is derived from a Greek word meaning:-
- **Schizo** - **split**
Phrenia – **mind** => meaning **split mind**
- Generally schizophrenia is a serious psychiatric disorder characterized by impaired communication with loss of contact with reality and deterioration from a previous level of functioning in work, social relations or self-care.

Schizophrenia....

- Schizophrenia is a **major psychiatric disorder** with varied presentation, which primarily **disturbances of**
 - **Thought**
 - **Perception**
 - **Affect & mood**
 - **Behavior and reality testing**
- Begins **before age 25**, persists **throughout life**, and affects persons of **all social classes**

Epidemiology

- **Prevalence**

- Prevalent in **all cultures and religions**.
- **Life time** prevalence rate is 1%

- **Age and sex**

- **M = F**
- Men have **earlier age of onset**, around 15 to 25yrs
- Women have **age of onset between 25 & 35yrs**
- **Rare** in less than 10yr and **greater than 60 yr**

Etiology of Schizophrenia

There are different etiologies such as :-

- Genetic
- Neuron degenerative hypothesis
- Neuron developmental hypothesis
- Excitotoxicities
- Biochemical hypothesis

Genetic factors

- The **likelihood** of a person having schizophrenia is **correlated with the closeness** of the relationship to an affected relative
- **Family studies** – relative **risk to sibling 12%**
- **Twin studies** – MZ: DZ concordance
50%: 12%
- **Adoption studies** - increased risk in biological (but not adoptive) relatives

Neurodevelopmental Model

- Neurodevelopmental model **supposes** in schizophrenia the presence of **“silent lesion”** in the brain, mostly in the parts, important for the **development** of integration
 - Frontal,
 - Parietal and
 - Temporal,
- which is caused by different factors (**genetic, inborn, infection, trauma...**) during very early development of the brain in prenatal or early postnatal period of life.

Neurodevelopmental Hypotheses of Schizophrenia

- the possibility that schizophrenia could have a Neurodevelopmental basis includes
 - observations that schizophrenia is **increased** in those with:
 - **Fetal history of obstetric complications** ranging from
 - **Viral infections**
 - **Starvation**
 - **Autoimmune processes** and
 - other such problems in the pregnant mother.

Neurodegenerative Hypotheses of Schizophrenia

- The presence of both **functional** and **structural** abnormalities demonstrated in neuroimaging studies of the brain of schizophrenics
- It suggests that a **neurodegenerative process** with progressive **loss of neuronal function** may be ongoing during the course of the disease.
- It is also suggested by the progressive nature of the course of illness in schizophrenia

- One major idea proposed to explain the downhill course of schizophrenia and :
 - the development of treatment resistance
 - is that neurodegenerative events in schizophrenia
 - may be mediated by a type of **excessive action of the neurotransmitter glutamate** that is **excitotoxicity**.
- The excitotoxic hypothesis of schizophrenia proposes that **neurons degenerate** because of **excessive excitatory neurotransmission at glutamate neurons**

Schizophrenia...

- **Medical illness**
 - ~80% co-occurring, 50% untreated
 - Higher mortality rate from accidents and natural causes than the general population
- **Infection & season of birth**
 - Winter birth
 - Second trimester viral influenza infection(maternal)
- **Substance abuse**
 - 50% rate of substance abuse other than nicotine
 - 90% nicotine, ↓EPS & cognitive side effect,?↓hallucination

Biochemical Factors

- **Dopamine Hypothesis**
 - schizophrenia results from **too much dopaminergic activity**. The theory evolved from two observations.
 - **Excessive dopamine release** in patients with schizophrenia has been **linked to the severity of positive** psychotic symptoms
- **Serotonin Hypothesis**
 - **Serotonin excess** as a cause of both **positive and negative symptoms** in schizophrenia
 - serotonin antagonist activity of second-generation antipsychotics, leading to **decreased positive symptoms**

Biochemical Factors cont'd

Norepinephrine. **Anhedonia-** the impaired capacity for emotional gratification and the **decreased ability to experience pleasure** has long been noted to be a prominent feature of schizophrenia.

GABA - has been implicated in the **pathophysiology** of schizophrenia

- Some schizophrenics have a **loss of GABAergic neurons in the hippocampus.**
- **GABA** has a **regulatory effect on dopamine activity**, and the **loss of inhibitory GABAergic neurons** could lead to the **hyperactivity of dopaminergic neurons.**

Clinical Symptoms and Diagnostic Characteristics

- Eugene Bleuler introduced the term schizophrenia and
- He cited symptoms referred to as **Bleuler's 4 A's** of schizophrenia :
 - **Affective disturbance**
 - **Autistic thinking** (turning towards self)
 - **Ambivalence** (conflict of ideas) and
 - **Association looseness** (incoherence)

Clinical Symptoms...

- **Affective disturbance** refers to the person's **inability** to show appropriate emotional responses.
- **Autistic thinking** is a thought process in which the individual is **unable to relate to others or to the environment**.
- **Ambivalence** refers to **contradictory or opposing** emotions, attitudes, ideas, or desires for the same person, thing, or situation.
- **Association Looseness** is the **inability to think logically**. Ideas expressed have little, if any, connection and shift from one subject to another

Symptoms in schizophrenia

Positive symptoms

- **Hallucination**
- **Delusion**
- **Disorganized speech**
- **Grossly disorganized behavior**
- **Catatonic behavior**

Negative symptoms (5A)

- **Anhedonia**
- **Affective flattening**
- **Alogia** (speech abnormalities-)
- **Apathy**(avolitions) (impaired grooming and hygiene)
- **Attentions** impairment

Other symptoms

- **Cognitive symptoms**
 - Attention deficit
 - Memory problem
 - Problem in information processing
- **Mood symptoms**
 - Anhedonia

DSM-V-TR Diagnostic criteria of schizophrenia

- A. Two or more of the following**, each present for significant portion of time during **one month period**. At least one of these must be (a),(b),(c)
 - A. Delusion**
 - B. Hallucination**
 - C. Disorganized speech**
 - D. Grossly disorganized or catatonic behavior**
 - E. Negative symptoms**
- B. Social/Occupational dysfunction:-** work, interpersonal relation or self-care
- C. Duration:-** 6 months or more
- D. Exclude** Schizoaffective and mood
- E. Exclude** Substance/general medical condition
- F. Relationship** to pervasive developmental disorder.



Phases of schizophrenia

Three phases

Prodromal phase

- ✓ Social withdrawal
- ✓ Subtle changes in behavior- aloof, odd, absent-minded
- ✓ Months to year

Active phase

- ✓ Predominant psychotic symptoms- delusion, hallucination
- ✓ Disorganized behavior/speech

Residual phase

- Positive symptoms absent or no longer prominent.
- negative symptoms prominent

Classification of Subtypes of Schizophrenia

Paranoid:

- Preoccupation with one or more delusions or frequent auditory hallucinations
- None of the following is prominent: disorganized speech, disorganized or catatonic behavior, or flat or inappropriate affect

Classification...

Catatonic: At least two of the following are present:

- **Motor immobility** (ie. rigidity), waxy flexibility, or stupor
- **Excessive motor activity** that is purposeless
- Extreme **negativism** or **mutism**
- **Peculiarities of voluntary movement** as evidenced by posturing, stereotyped movements, prominent mannerisms or prominent grimacing
- **Echolalia** (repeats all words or phrases heard) or **echopraxia** (mimics actions of others)

Classification...

Disorganized:

- All of the following are prominent and criteria are not met for catatonic type:
- Disorganized speech
- Disorganized behavior
- Flat or inappropriate affect

Classification...

Undifferentiated:

- Meets diagnostic characteristics but not the criteria for paranoid, disorganized, or catatonic subtypes

Residual:

- Absence of prominent delusions, hallucinations, disorganized speech, and grossly disorganized or catatonic behavior
- Continuing evidence of, in attenuated form, the presence of negative symptoms or two or more symptoms of diagnostic characteristic

Clinical Example

- **The Client With Schizophrenia, Paranoid Type**
- BW, a 35-year-old mechanic, was brought to the admissions office by his wife because he had exhibited **strange behavior for several months.**
- He **accused** his **wife of poisoning his food,** spending all his **money,** having an **affair with his boss,** and **telling stories about him**

Clinical Example...

The Client With Schizophrenia, Catatonic Type

- CS, a 25-year-old engineer, was **admitted to the hospital as result of dehydration because of refusing to eat.**
- During his hospitalization, CS was **negativistic, refusing nursing care, food, and medication.**
- He **rarely spoke** and **assumed uncomfortable positions in bed for long periods.**
- When **placed in various positions** by the nurse during the morning bath or shower, CS **remained in the positions until the nurse changed them.**
- He also exhibited **purposeless movements of his hands and feet** while sitting in a chair.

Clinical Example...

- He displayed **no facial expressions** during his initial interview and **became quite argumentative** when questioned about his job
- At the end of the interview, BWV confided in the interviewer that **he had been receiving messages from Jesus Christ** while **watching television.**

Clinical Example...

- **The Client With Schizophrenia, Disorganized Type**
- MJ, a **19-year-old waitress**, was seen in the admitting office of a psychiatric hospital.
- During the initial interview, **she giggled inappropriately.**
- Her **long, uncombed hair fell over her face, concealing her facial expressions.**

Clinical Example...

- She **mumbled incoherently** at times and displayed the **behavior** of a **13- or 14-year-old adolescent**.
- She **complained** of numerous **aches and pains** and stated that **voices told her she was being punished** for not cleaning her room.
- MJ's mother stated that she **remained in her room at home** and **did not socialize** with friends.
- Her parents sought help when they noticed her behavior regressing during the past 2 months

Differential diagnosis

- Organic mental disorder
- Major affective disorder
- Schizophreniform disorder
- Paranoid disorder
- Mental retardation
- The developmental struggle of adolescent

Prognosis:

- Prognosis depends on **onset, stress, pre morbid personality** and **social and economic status**

Treatment

- **Duration therapy**
 - **1st episode** maintenance therapy for **1-2 years**
 - **Multiepisode** maintenance therapy for **5 years**
 - **Several severe episodes** ,suicidality , violence, aggression may require **indefinite** maintenance therapy

Outcome of schizophrenia

Devastating illness

- >50% of patients- long-term incapacity
- 20-30% to lead somewhat normal life
- 20-50% attempt suicide
- >10% of patients commit suicide

Management

- Hospitalization
- Pharmacotherapy
- Psychosocial treatment
- Rehabilitation & social integration

Antipsychotics

- Divided into two groups:

1. Typical

(1st generation)(DRA)

- Chlorpromazine } low potent
- Thioridazine }
- Fluphenazine } high potent
- Trifluoperazine }
- **Haloperidol** }

- Side effect

- EPS

1. Atypical

(2nd generation)(DSRA)

- **Risperidone 4-6mg**
- **Olanzapine 5-15mg**
- Clozapine
- Quetiapine
- Ziprasidone

- Side effect

- Weight gain

Extrapyramidal effects (EPS)

- Extrapyramidal symptoms are most commonly caused by typical **antipsychotic drugs** that **antagonize dopamine D2 receptors**.
- The **most common typical** antipsychotics associated with EPS are **haloperidol** and **fluphenazine**.
- **Examples of extrapyramidal effects include:**
 - **Akathisia**: Feeling **restless** like inability sit still. ...
 - **Dystonia**: involuntarily muscles contract. ...
 - **Parkinsonism**: Symptoms are similar to Parkinson's disease. .
 - **Tardive dyskinesia**: Facial movements happen involuntarily

Choice of antipsychotic

A . Clinical factors

- Patients' prior experience
- Clinical response
- Subjective response
- Sensitivity to EPS

B. Non clinical factors

Cost ,availability ,uniformity of supply

Initial work up

- In **emergency** setting with the exception of clozapine clinicians **can administer without conducting** a physical or neurological exam.
- **Usual assessment**
 - Complete blood count
 - Liver function test
 - ECG

Contraindication

- History of severe **allergic reaction**
- The possibility that a **patient has ingested substance that will interact with antipsychotic** to induce CNS depression
- Severe **cardiac abnormality**
- A high **risk of seizure**

- Route of administration of acute phase treatment
 - P.O if cooperative
 - I.M or I.V if not

Antipsychotic medications

MEDICATION	HALOPERDIOL	CHLORPROMAZINE
Starting dose	1.5 – 3 mg	75 mg
Typical effective dose	3 – 20 mg/day	75 – 300 mg/day
Route	Oral/i.m.	Oral

Antipsychotic medications...

Antipsychotics(atypical)	Dose (mg)
Clozapin	300-450
Olanzapine	5-20
Resperidone	0.5-6
Ziprasidone	40-160 bid

Psychological

Support

Education / information

Behavioural (eg distraction for hallucinations)

Cognitive behavioural therapy (CBT) for delusions and hallucinations

Family therapy (expressed emotions)

Social

Housing

Finances

Employment / occupational therapy

Support groups (eg National Schizophrenia
Fellowship)

Stigma

- Main stigma surrounds issue of **violence**
- Public perception – **70% believe** that people with **schizophrenia are dangerous**
- **Media coverage** of mental illness – 77% associated **with aggression**
- **Facts:** aggression **more common** than in **general population** but much less than that due to alcohol/drug intoxication.
- **Serious violence rare** (<5%) and >95% homicides committed by people **without schizophrenia**

Marked extra pyramidal side effects (EPS)

Parkinson's syndrome—characterized by :

- Rigidity
- Mask-like faces
- Bradykinesia
- Tremor, and
- Stooped posture



Extra pyramidal side effects...

- **Dystonia** - sustained contraction of the muscles of neck, eyes, tongue, jaw and other muscle groups, typically occurring within 3-5 days after initiation of the neuroleptic. Dystonia's are often very painful and frightening to patients.
- **Akathasia** – characterized by strong feelings of inner restlessness, which are manifest by difficulty remaining still and excessive walking or pacing.

CONT..

- **Tardive dyskinesia** is an **involuntary movement** disorder involving the tongue, mouth, fingers, toes, and other body parts.
- It is characterized by **chewing movements, smacking and licking of the lips, sucking movements, tongue protrusion, blinking, grimaces and spastic facial distortions.**

Treatment of acute psychotic stage

Rapid loading – no advantage

- Start low, go slow- slowly increase dosage
- If no response in 3-8 weeks change to another drug

Highly agitated patient

- Haloperidol parenteral Q 30- 120 min, until agitation subsides OR
- Haloperidol + Benzodiazepine – parenteral Q 30 min until agitation subsides

Electroconvulsive therapy (ECT)

For acute and subacute forms of schizophrenia

- Patients **not responding** to antipsychotic medications
- **Severe catatonic symptoms** – stupor, extreme agitation
- **Severe depression** secondary to schizophrenia

Electroconvulsive therapy (ECT)



Nursing interventions

- Nursing interventions focus on assisting the patient to meet the following **goals**:
- 1. Establish a **trusting relationship**;
- 2. Alleviate **anxiety**;
- 3. Maintain **biological integrity**; and
- 4. **Establish** clear, consistent, and open **communication**

Nursing interventions...

- The nurse must **establish** a **therapeutic relationship** so that effective communication with the patient can take place
- The nurse must remember that **all behavior is meaningful** to the patient, if not to anyone else

Nursing interventions...

- **Reality should be presented** when caring for the patient who is disoriented.
- The nurse can do this by **pointing out what would be appropriate behavior**, for instance, 'I'd like you to put your shoes on now'
- **Recognizing** the presence of hallucinations and delusion, but **not reinforcing such behavior** or thoughts is an appropriate response when interacting with patients.
- The nurse should **look for factors causing hallucinations** and attempt to intervene before they occur.

Nursing interventions...

- Safety measures may need to be incorporated to protect the patient who displays poor judgment, disorientation, destructive behavior, suicidal ideation, or agitation.
- Limit setting, acknowledging spatial territory giving the patient room to breathe' and providing protective safety measures are examples of such nursing interventions.
- The patient must be protected from her or himself because she or he may injure her or himself accidentally, or may try to destroy her or himself or attack other patients as a result of auditory hallucinations or paranoid ideations.

Nursing interventions...

- Efforts should be made **to plan activates** to increase the patient's self concept
- **Sincere compliments** should be given as often as possible, focusing on positive aspects of the person's personality or capabilities
- Encourage **participation in activates**

Nursing interventions...

- The nurse must observe for **extra pyramidal side effects of psychotropic drugs** and **monitor the patient's willingness** to take the drugs
- Patients **may refuse to take medication, pretend to take medication by palming it, or pretend to swallow the medication while retaining the pill in the mouth** (only to get rid of it at the first possible moment)

Nursing diagnoses and interventions for patients with schizophrenic disorders

Nursing diagnoses	Nursing Interventions
Sensory-perceptual alteration: Disoriented to place and person, disoriented in time.	Call the patient by name. Present reality when talking to or working with the patient. Keep a calendar in clear view to orient the patient daily. Provide a protective, safe environment
Social isolation: Withdrawal	Assign one member of the health care team to establish a one-to-one relationship. Provide a structured list of activities such as times to awaken, shower, and eat. Spend a specific amount of time daily with the patient. Set limits regarding amount of times spent alone in room
Alteration in thought process: Delusional	Present reality when talking to or working with the patient. Ignore the delusion but do not attempt to disprove it or argue with the patient. Set limits by instructing the patient not to discuss the delusion with others

Alteration in thought process: Hallucinations	Decrease environmental stimuli such as loud music or television shows, extremely bright colors, or flashing lights. Present reality, for example: “The voices may be real to you but I don’t hear anything.” Attempt to identify precipitating factors by asking the patient what happened before the onset of the hallucination
Ineffective individual coping: Regression	Assess the patient’s present developmental level. State expected behavior to the patient. Set limits to discourage regressive behavior.
Agitation and unpredictable behavior	Recognize signs of increasing agitation. Decrease environmental stimuli that could be upsetting to the patient
Sensory-perceptual alteration: Suspiciousness	Be sincere and honest when talking with the patient. Avoid making promises that can not be fulfilled. Face the patient while talking. Avoid whispering or any other behavior that may cause the patient to feel that you are talking about him. Give detailed explanations of tests, procedures, and so forth to the patient. Allow the patient to help to prepare food or have food brought from home if he or she refuses to eat (because s/he thinks food is poisoned).

Quiz

- 1. Ato Bekele a 30 years old teacher comes to psychiatric clinic presented with positive symptoms of schizophrenia : overt psychotic or distorted behavior, hallucinations, delusions, or suspiciousness
- According to the biochemical theory schizophrenia such behaviours of Ato Kebede are said to be caused by excessive amounts of :-
 - A.Serotonin
 - B.Biogenicamines
 - C.Dopamine
 - D.Epinephrine

Quiz...

2. Ato Bekele was diagnosed as having schizophrenia

Which of the following drugs is used for the treatment of Ato Kebede ?

A. Chlorpromazine

B. Diazepam

C. Morphine

D. Levodopa

3. Motor immobility such as waxy flexibility or stupor is observed in patients with ____ schizophrenia

- A. Paranoid
- B. Disorganized
- C. Catatonic
- D. Residual

4. Schizophrenia currently presents a crisis situation. Approximately 1% of the earth's population, suffer from schizophrenia or schizophrenic-like disorders.

Schizophrenia is a _____ disorder:-

- A. Mood
- B. Neurotic
- C. Psychotic
- D. Anxiety

Quiz...

5. Obo Tolessa is a 38 years old engineer and was brought to hospital because he exhibits strange behavior and he is accusing his wife of poisoning his food and having an affair with his boss.

He is argumentative and claimed that he had been receiving message from God while watching television.

Based on these symptoms your probable diagnosis of Ato kebede will be schizophrenia of _____ type

- A. Disorganized
- B. Catatonic
- C. Residual
- D. Paranoid

6. Sara is , an 18 -year-old college student , was seen in the admitting office of a psychiatric hospital. During the initial interview, she giggled inappropriately. Her long, uncombed hair fell over her face, concealing her facial expressions. She complained of numerous aches and pains and stated that voices told her she was being punished for not cleaning her room. Her mother stated that she remained in her room at home and did not socialize with friends. She is acting like a 6 years old child.

Based on the above signs and symptoms, the probable diagnosis of Sara is _____schizophrenia.

- A. Disorganized
- B. Catatonic
- C. Residual
- D. Paranoid

7. A serious mental disorder characterized by impaired communication with loss of contact with reality and deterioration from a previous level of functioning in work, social relationships, or self care is :-

- A. Paranoid disorder
- B. Schizophrenia
- C. Mania
- D. Depression

8. Obo Tolessa was diagnosed as having
paranoid type schizophrenia .

All the following are clinical features of paranoid type
schizophrenia except :-

- A. Anger
- B. Fearfulness
- C. Violence
- D. Sad face

Quiz...

9. Obo Yusuf was admitted to psychiatric ward diagnosed as having pchizophrnia. Which one of the following nursing diagnosis will be appropriate for a client with schizophrenia ?

- A. Alternation in bowel elimination: Constipation
- B. Alteration in self- concept
- C. Potential for injury: Suicidal ideation
- D. Alteration in thought process: Delusional

10. Which nursing response would be most appropriate when a client talks about hearing voices?

A. I do not hear the voices that you say you hear.

B. Those voices will disappear as soon as the medicine works

C. Try to think about positive things instead of voices.

D. Voices are only in your imagination

The End

